

LLM2CLIP: Powerful Language Model Unlocks Richer Visual Representation

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Abstract

Contrastive Language-Image Pre-training (CLIP) has achieved remarkable success in vision-language tasks, yet its text encoder—with only about one third the parameters of the vision encoder—limits its capacity for deep language understanding. In this work, we investigate LLM2CLIP, a framework that replaces CLIP’s original text encoder with a large language model (LLM) to unlock richer visual representation. Through a two-stage training pipeline involving LLM caption-contrastive fine-tuning followed by vision encoder adaptation, LLM2CLIP achieves substantial improvements in image-text retrieval. We conduct three sets of experiments: (1) a retrieval comparison on Flickr30k showing that LLM2CLIP outperforms CLIP by +8.6 on image-to-text and +17.7 on text-to-image retrieval; (2) an adaptor design study demonstrating that a lightweight linear adaptor converges more efficiently than a multi-layer MLP under small-scale training; and (3) a comprehensive robustness evaluation under text perturbations (semantic distractors, object swaps, and color/spatial swaps) on MSCOCO 2014, revealing that LLM2CLIP is more tolerant to irrelevant semantic noise while being more sensitive to meaningful visual-semantic changes—precisely the behavior desired of an ideal vision-language model.

1 Introduction

Contrastive Language-Image Pre-training (CLIP) (Radford et al., 2021) has emerged as a foundational model for vision-language tasks, learning to align visual and textual representations through large-scale image-text pairs. However, a fundamental limitation of CLIP lies in the asymmetry between its encoders: the text encoder possesses only about one third of the parameters of the vision encoder, constraining its ability to capture nuanced semantic information from textual descriptions.

Large Language Models (LLMs), on the other hand, have demonstrated exceptional language understanding capabilities and open-world knowledge. This motivates the central question explored in this work: *Can we leverage the rich semantic understanding of LLMs to enhance CLIP’s visual representation?*

LLM2CLIP (Huang et al., 2024) proposes an elegant solution by replacing CLIP’s lightweight text encoder with a fine-tuned LLM, enabling stronger image-text alignment through richer textual features. In this paper, we present a comprehensive study of the LLM2CLIP framework, organized around three research questions:

1. Does LLM2CLIP improve image-text alignment compared with the original CLIP?
2. How does the adaptor design affect downstream performance?
3. Is LLM2CLIP more robust to various types of text perturbations?

We conduct experiments on image-text retrieval (Flickr30k), adaptor architecture comparison, and a detailed robustness analysis on MSCOCO 2014 with three types of semantic perturbations: semantic distractors, object swaps, and color/spatial swaps. Our findings demonstrate that LLM2CLIP not only achieves superior retrieval performance but also exhibits a desirable robustness profile—it is tolerant to irrelevant noise while remaining highly sensitive to meaningful visual-semantic changes.

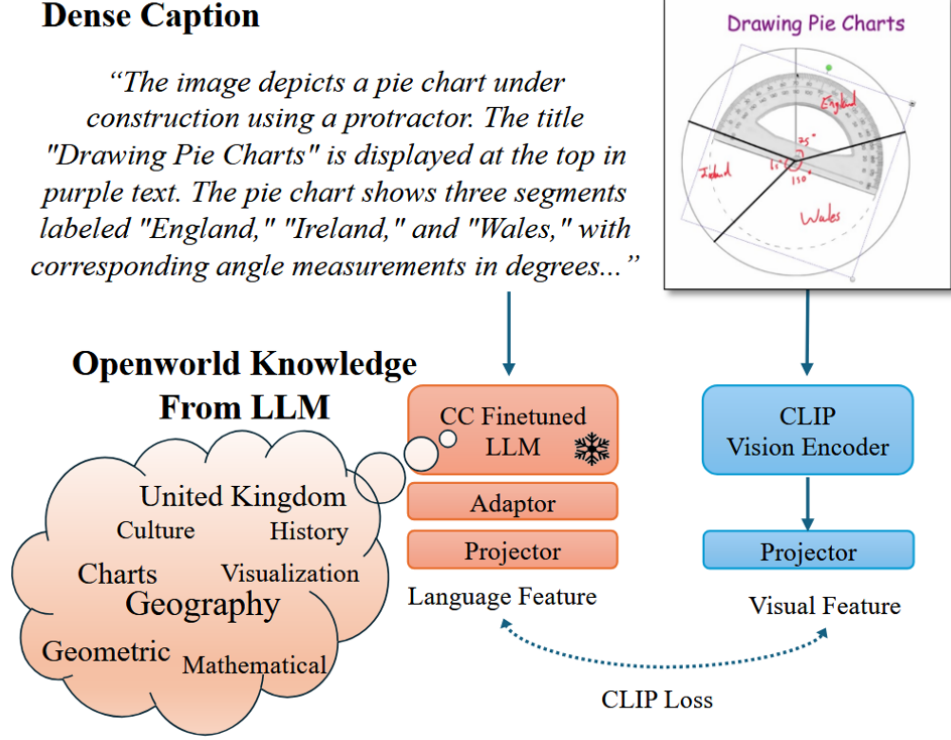


Figure 1: The LLM2CLIP framework. The LLM (e.g., Llama-3-8B) is first fine-tuned into a caption embedding model via LoRA, average pooling, bidirectional attention, and supervised SimCSE (Stage 1). The fine-tuned LLM then replaces the original CLIP text encoder, and a lightweight adaptor aligns the LLM text embeddings with the CLIP-compatible feature space while the vision encoder and adaptor are fine-tuned (Stage 2).

2 Background: Why Does CLIP Need an LLM?

CLIP (Radford et al., 2021) employs a dual-encoder architecture, consisting of a vision encoder and a text encoder, trained contrastively on a large collection of image-text pairs. While this approach has proven highly effective, the text encoder is significantly smaller than the vision encoder—roughly one third of its parameter count. This architectural asymmetry limits the model’s capacity for fine-grained semantic understanding.

In contrast, modern LLMs such as Llama-3-8B possess substantially stronger language understanding and open-world knowledge. These models can capture nuanced semantic relationships, understand complex textual descriptions, and maintain robust representations even under textual variations. By integrating an LLM into the CLIP framework, we can potentially bridge the semantic gap between visual and textual modalities.

Figure ?? illustrates the contrastive pre-training paradigm of CLIP and highlights the key motivation: the LLM provides richer language features that can better guide the visual representation learning process.

3 Method Overview: Two-Stage Framework

The LLM2CLIP framework employs a two-stage training pipeline, as illustrated in Figure 1.

3.1 Stage 1: Converting the LLM into a Caption Embedding Model

In the first stage, a pre-trained LLM (e.g., Llama-3-8B) is transformed into an effective caption embedding model following the LLM2Vec approach (BehnamGhader et al., 2024). This involves several key modifications:

- **LoRA fine-tuning:** Low-rank adaptation is applied to efficiently fine-tune the LLM for the caption embedding task while preserving its pre-trained knowledge.
- **Average pooling:** Instead of using only the last token representation, average pooling across all tokens is employed to produce a comprehensive sentence embedding.
- **Bidirectional attention:** Although LLMs are inherently causal (unidirectional), bidirectional attention is introduced to allow the model to capture full contextual information from both directions of the input caption.
- **Supervised SimCSE:** A supervised contrastive learning objective (SimCSE) is applied using caption pairs to train the model to produce discriminative embeddings that capture semantic similarity.

3.2 Stage 2: Vision-Language Alignment with Adaptor

In the second stage, the fine-tuned LLM replaces the original CLIP text encoder:

- **Freeze the LLM:** The LLM parameters are frozen to preserve the rich semantic representations learned in Stage 1.
- **Lightweight adaptor:** A trainable adaptor module is added to map the LLM’s high-dimensional text embeddings (e.g., 4096-d from Llama-3-8B) into the CLIP-compatible feature space (e.g., 512-d or 768-d).
- **Fine-tune vision encoder and adaptor:** Only the vision encoder and the adaptor are fine-tuned using the standard CLIP contrastive loss, enabling efficient alignment between the visual and enhanced textual representations.

The adaptor serves as a critical bridge between the LLM and the vision encoder. Its design choice significantly impacts both training efficiency and downstream performance, as we investigate in our experiments.

4 Experiments

We conduct three sets of experiments to comprehensively evaluate the LLM2CLIP framework.

4.1 Experiment 1: Image-Text Retrieval Comparison

4.1.1 Setup

We compare the zero-shot image-text retrieval performance of CLIP and LLM2CLIP on the Flickr30k dataset. The task has two retrieval directions:

- **Image-to-Text (I2T):** Given an image, retrieve the most relevant caption from the candidate pool.
- **Text-to-Image (T2I):** Given a caption, retrieve the most relevant image from the candidate pool.

Both models use the ViT-B/16 vision encoder architecture. The CLIP model uses its standard text encoder, while LLM2CLIP replaces it with the fine-tuned LLM and a trainable adaptor.

Model	I2T	T2I
CLIP ViT-B/16	81.7	63.5
CLIP ViT-B/16 + LLM2CLIP	90.3	81.2

Table 1: Zero-shot image-text retrieval results on Flickr30k (1K test set). I2T and T2I denote image-to-text and text-to-image retrieval accuracy, respectively.

Task	Default Adaptor	Ours (Linear)	Δ
T2I R@1	25.7	36.0	+10.3
T2I R@5	57.4	68.1	+10.7
T2I R@10	71.5	80.2	+8.7
I2T R@1	52.2	59.2	+7.0
I2T R@5	81.7	86.7	+5.0
I2T R@10	89.9	93.6	+3.7

Table 2: Image-text retrieval results on Flickr30k (1K test set) comparing two adaptor architectures. Both models use OpenAI CLIP ViT-B/16 as the vision encoder and LLM2Vec + Llama-3-8B as the text encoder, trained on 5,000 images for 5 epochs.

4.1.2 Results

The results are presented in Table 1. LLM2CLIP achieves substantial improvements over the baseline CLIP model across both retrieval directions, with a particularly notable gain of +17.7 points on text-to-image retrieval. This demonstrates that the richer semantic representations from the LLM significantly enhance the alignment between textual and visual features.

4.2 Experiment 2: Adaptor Architecture Study

4.2.1 Setup

We investigate how the adaptor design affects training convergence and downstream performance. We compare two adaptor architectures:

- **Default (4-layer MLP):** $L2N \rightarrow 4 \times (LN \rightarrow \text{Linear} \rightarrow \text{GELU} \rightarrow \text{Linear}) + \text{Res} \rightarrow LN \rightarrow \text{Linear}$
- **Simplified (Linear):** $L2N \rightarrow LN \rightarrow \text{Linear}$

Both adaptors map the 4096-dimensional LLM output (from LLM2Vec + Llama-3-8B) into the 512-dimensional CLIP feature space. We train on the Flickr30k training set (5,000 images, 5 captions each) for 5 epochs.

4.2.2 Results

The retrieval results are presented in Table 2. Under small-scale training (5,000 images, 5 epochs), the simplified linear adaptor (Ours) consistently outperforms the default 4-layer MLP across all recall metrics in both retrieval directions. Notably, the linear adaptor achieves a substantial improvement on T2I R@1 (+10.3 points), suggesting that it learns a more effective mapping from the LLM embedding space to the CLIP feature space when training data is limited. The multi-layer MLP is more prone to overfitting and optimization difficulty under small-scale training conditions, whereas the lightweight linear adaptor converges more easily and generalizes better.

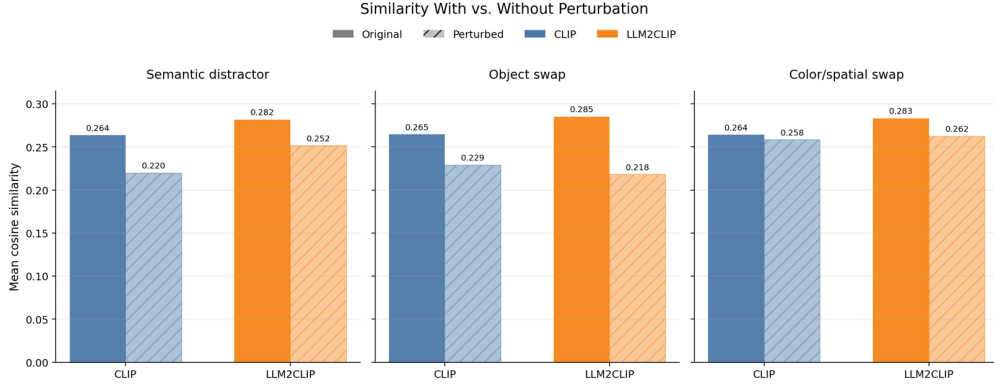


Figure 2: Mean cosine similarity comparison between CLIP and LLM2CLIP under three perturbation types. For each perturbation, both the original (unperturbed) and perturbed similarity scores are shown. LLM2CLIP consistently achieves higher original similarity and exhibits larger drops under object swap and color/spatial swap, while showing smaller degradation under semantic distractors.

4.3 Experiment 3: Text Perturbation Robustness

4.3.1 Setup

To comprehensively evaluate the robustness of LLM2CLIP versus CLIP, we design a text perturbation experiment on the MSCOCO 2014 dataset (5K test set). We compare two models:

- **CLIP:** openai/clip-vit-large-patch14-336
- **LLM2CLIP:** microsoft/LLM2CLIP-Openai-L-14-336 with
microsoft/LLM2CLIP-Llama-3-8B-Instruct-CC-Finetuned

For each image-caption pair, we compute:

$$s_{\text{original}} = \cos(\mathbf{e}_{\text{image}}, \mathbf{e}_{\text{text}}^{\text{original}}) \quad (1)$$

$$s_{\text{perturbed}} = \cos(\mathbf{e}_{\text{image}}, \mathbf{e}_{\text{text}}^{\text{perturbed}}) \quad (2)$$

$$\delta = s_{\text{original}} - s_{\text{perturbed}} \quad (3)$$

A larger δ indicates that the model is more sensitive to the perturbation. We evaluate three types of text perturbations:

- **Semantic distractor** (25,010 pairs): An unrelated caption from another image is appended to the original caption using various templates (e.g., "Target image description: [original]. Unrelated description: [distractor]").
- **Object swap** (14,233 pairs): Core object nouns in the caption are replaced with semantically different ones (e.g., "man" → "woman", "girl" → "boy").
- **Color/spatial swap** (18,131 pairs): Color adjectives or spatial prepositions are swapped (e.g., "red" → "blue", "on" → "under").

4.3.2 Results

The quantitative results are visualized in Figures 2 and 3. Figure 2 shows the mean cosine similarity before and after perturbation, and Figure 3 presents the distribution of δ values across perturbation types.

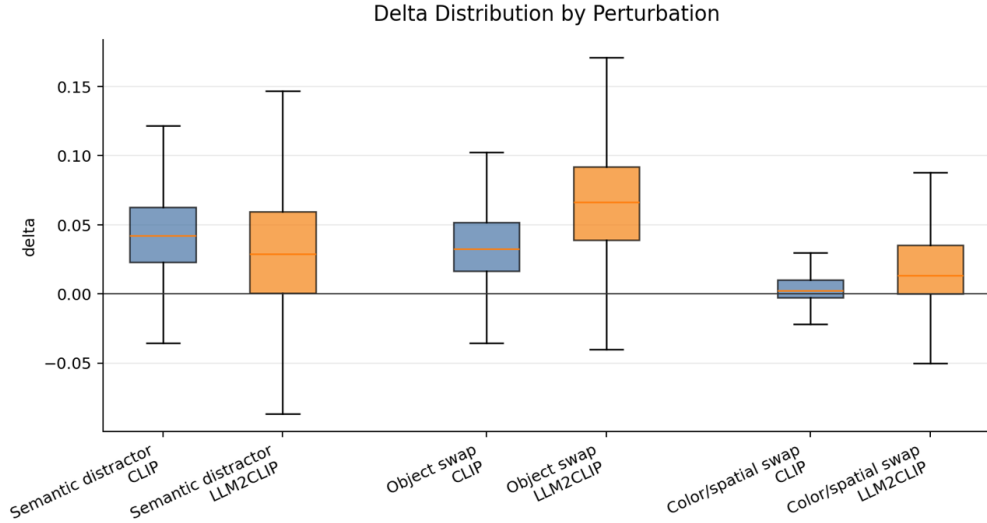


Figure 3: Distribution of δ (similarity drop) by perturbation type for CLIP and LLM2CLIP. LLM2CLIP shows a wider and more positive distribution under object swap and color/spatial swap, indicating stronger sensitivity to fine-grained semantic changes. Under semantic distractors, LLM2CLIP exhibits a smaller and more variable distribution, suggesting better tolerance to irrelevant noise.

4.3.3 Analysis

We identify three key findings from the robustness analysis:

Higher baseline alignment. LLM2CLIP achieves a higher mean cosine similarity on original (unperturbed) captions (0.2817 vs. 0.2635 for CLIP), confirming that the LLM-enhanced text encoder provides stronger image-text alignment in clean conditions.

Tolerance to semantic distractors. Under semantic distractors, CLIP shows a larger similarity drop ($\delta_{\text{mean}} = 0.0439$) compared to LLM2CLIP ($\delta_{\text{mean}} = 0.0302$). Furthermore, 93.9% of CLIP samples exhibit a positive δ , while only 75.0% of LLM2CLIP samples do. This indicates that LLM2CLIP is better at maintaining relevant image-text similarity even when exposed to extraneous descriptions. We hypothesize that the LLM’s strong language understanding enables it to leverage template cues (e.g., “unrelated description,” “ignore the following”) to effectively filter out irrelevant information.

Sensitivity to meaningful semantic changes. Under object swaps, LLM2CLIP exhibits a substantially larger similarity drop ($\delta_{\text{mean}} = 0.0671$, approximately $1.89\times$ that of CLIP at 0.0356). Similarly, under color/spatial swaps, LLM2CLIP’s $\delta_{\text{mean}} = 0.0210$ is approximately $3.67\times$ that of CLIP at 0.0057. This demonstrates that LLM2CLIP is significantly more sensitive to fine-grained semantic mismatches—precisely the capability needed for accurate vision-language understanding.

Summary of robustness profile. The combination of (a) tolerance to irrelevant noise and (b) sensitivity to meaningful semantic changes represents an ideal robustness profile for a vision-language model. LLM2CLIP achieves this desirable balance, whereas CLIP tends to be uniformly affected by both types of perturbations.

5 Conclusion

In this work, we have presented a comprehensive study of the LLM2CLIP framework, which enhances CLIP’s visual representation by replacing its text encoder with a fine-tuned large language model. Our key findings are:

1. **Superior retrieval performance:** LLM2CLIP achieves substantial improvements over CLIP on Flickr30k image-text retrieval (+8.6 on I2T, +17.7 on T2I), demonstrating that the richer semantic understanding from the LLM significantly enhances cross-modal alignment.
2. **Efficient adaptor design:** Under small-scale training conditions, a lightweight linear adaptor outperforms a 4-layer MLP across all recall metrics (+10.3 on T2I R@1, +7.0 on I2T R@1), offering a practical advantage for resource-constrained settings.
3. **Desirable robustness profile:** On MSCOCO 2014, LLM2CLIP exhibits a unique combination of tolerance to irrelevant semantic noise (smaller δ under semantic distractors) and heightened sensitivity to meaningful visual-semantic changes (larger δ under object and color/spatial swaps). This dual behavior is precisely what one would expect from an ideal vision-language model.

These results collectively demonstrate that strengthening the text encoder through LLM integration is a highly effective strategy for improving vision-language models. Future work may explore scaling to larger LLMs, applying LLM2CLIP to additional downstream tasks, and further investigating the interplay between adaptor design and training data scale.

References

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